

Name: _____		
Student ID: _____	Section: _____	

Instructions

- Verify that your copy has **all 13 pages**.
- Show all work where space is provided; partial credit may be awarded.
- Do not use calculators unless your instructor says otherwise.
- Total possible points: **245**.

Word problems involving the mean

1. For what value of x will 8 and x have the same mean (average) as 27 and 5? [5pts]
 - A. 1.5
 - B. 8
 - C. 24
 - D. 40
2. If 6 and x have the same mean (average) as 2, 4, and 24, what is the value of x ? [5pts]
 - A. 5
 - B. 10
 - C. 14
 - D. 36
3. A set of data consists of four scores: x_1 , x_2 , x_3 , and x_4 . If the mean of the four scores is 9, what is the value of x_3 when $x_1 = 5$, $x_2 = 7$, and $x_4 = 11$? [5pts]
 - A. 9
 - B. 13
 - C. 3
 - D. 12
4. For five algebra examinations, Maria has an average of 88. What must she score on the sixth test to bring her average up to exactly 90? [5pts]
 - A. 92
 - B. 94
 - C. 98
 - D. 100

5. The exact average of a set of six test scores is 92. Five of these scores are 90, 98, 96, 94, and 85. What is the other test score? [5pts]
- A. 92
 - B. 91
 - C. 89
 - D. 86
6. During each marking period, there are five tests. If Vanita needs a 65 average to pass this marking period and her first four grades are 60, 72, 55, and 80, what is the lowest score she can earn on the last test to have a passing average? [5pts]
- A. 58
 - B. 65
 - C. 80
 - D. 100
7. In his first three years coaching baseball at High Ridge High School, Coach Batty's team won 7 games the first year, 16 games the second year, and 4 games the third year. How many games does the team need to win in the fourth year so that the coach's average will be 10 wins per year? [5pts]
- A. 13
 - B. 10
 - C. 3
 - D. 9
8. This year, John played in 10 baseball games. In these games he had hit the ball 2, 3, 0, 1, 3, 2, 4, 0, 2, and 3 times. In the first 10 games he plays next year, John wants to increase his average (mean) hits per game by 0.5. What is the total number of hits John needs over the first 10 games next year to achieve his goal? [5pts]
- A. 5
 - B. 2
 - C. 20
 - D. 25

9. The mean (average) weight of three dogs is 38 pounds. One of the dogs, Sparky, weighs 46 pounds. The other two dogs, Eddie and Sandy, have the same weight. Find Eddie's weight. [5pts]
- A. 32 lb
B. 33 lb
C. 34 lb
D. 35 lb
10. The students in Woodland High School's meteorology class measured the noon temperature every school day for a week. Their readings for the first 4 days were Monday, 56° ; Tuesday, 72° ; Wednesday, 67° ; and Thursday, 61° . If the mean (average) temperature for the 5 days was exactly 63° , what was the temperature on Friday? [5pts]
11. TOP Electronics is a small business with five employees. The mean (average) weekly salary for the five employees is \$360. If the weekly salaries of four of the employees are \$340, \$340, \$345, and \$425, what is the salary of the fifth employee? [5pts]
12. On his first 5 biology tests, Bob received the following scores: 72, 86, 92, 63, and 77. What test score must Bob earn on his sixth test so that his average (mean score) for all six tests will be 80? Show how you arrived at your answer. [10pts]

13. Juan received scores of 82, 76, 93, and 80 on his first four chemistry tests of the year. His goal is to have an 86 average in chemistry for his first five tests. What score must he earn on the next test to achieve an average of exactly 86? [5pts]
- A. 95
 - B. 97
 - C. 99
 - D. 100
14. Judy needs a mean (average) score of 86 on four tests to earn a midterm grade of B. If the mean of her scores for the first three tests was 83, what is the lowest score on a 100-point scale that she can receive on the fourth test to have a midterm grade of B? [5pts]
- A. 93
 - B. 94
 - C. 95
 - D. 96
15. On the first six tests in her social studies course, Jerelyn's scores were 92, 78, 86, 92, 95, and 91. Determine the median and the mode of her scores. If Jerelyn took a seventh test and raised the mean of her scores exactly 1 point, what was her score on the seventh test? [10pts]

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16. Noj has the following test scores:

76, 84, 69, 74, 91.

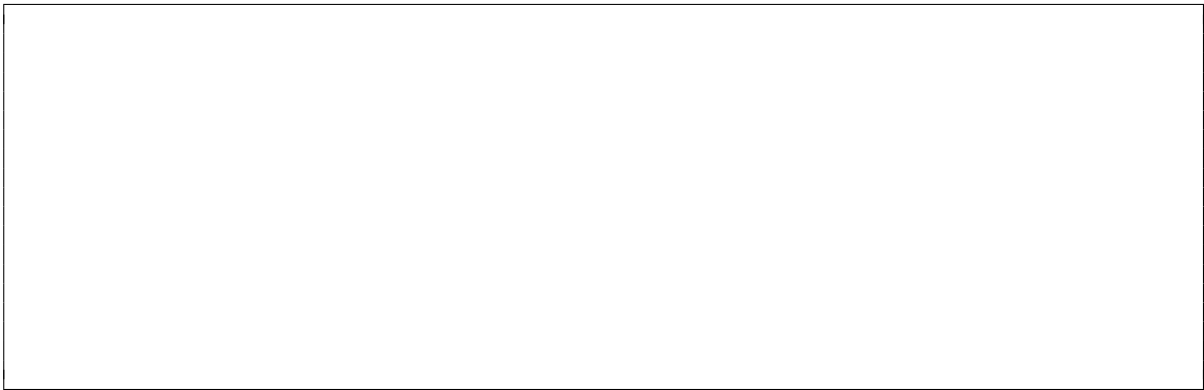
[10pts]

His teacher will allow him to retake the test on which he scored lowest. Noj wants an average of at least 82. Determine the least number of additional points Noj must score on the retest.

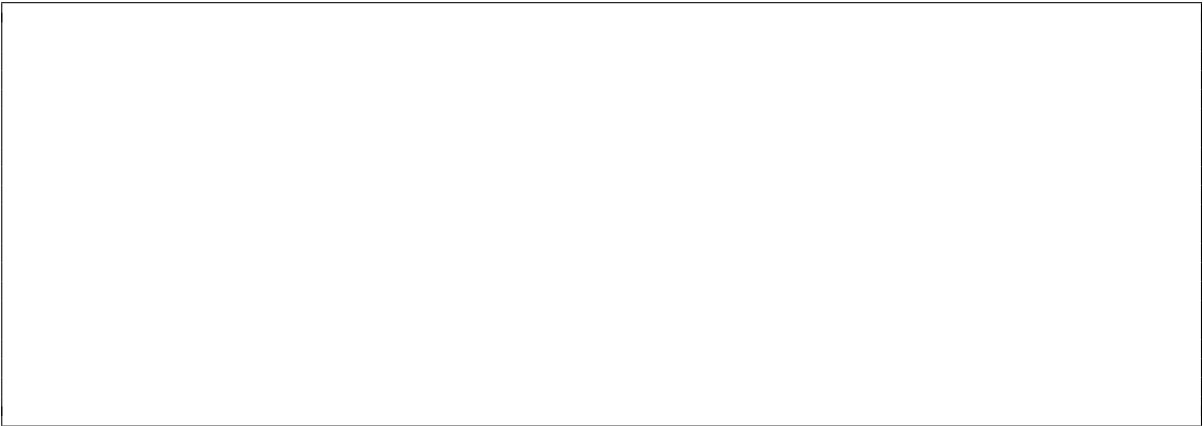


17. Tamika could not remember her scores from five mathematics tests. She did remember that the mean (average) was exactly 80, the median was 81, and the mode was 88. If all her scores were integers with 100 the highest score possible and 0 the lowest score possible, what was the lowest score she could have received on any one test?

[10pts]



18. Angelo, Brandon, and Carl work in the same office. Angelo's age is 4 years more than twice Carl's age. Brandon is 5 years younger than Carl. The average of the three ages is 41. Find the age of each of the men. [10pts]



19. The mean of three numbers is 25. The second number is four less than twice the first. The third number is two more than four times the first. Find the smallest number. [5pts]
- A. 10
 - B. 11
 - C. 12
 - D. 14

Word problems involving mean or median

20. Quinn's test scores are 95, 90, and 83. What must he score on his next test to have an average test score of 90 on the four tests? [5pts]

21. Desiree's test scores are 77, 90, 66, and 82. What score does she need on the last test in order to average 80 on her tests? [5pts]

22. Lawanda's test scores are 85, 97, 76, and 93. What score does she need on the last test in order to average 90 on her tests? [5pts]

23. Alicia's test scores are 77, 86, 66, and 80. What score does she need on the last test in order to average 80 on her tests? [5pts]

24. Lakita's test scores are 87, 96, 81, and 89. What score does she need on the last test in order to average 90 on her tests? [5pts]

25. Aisha's test scores are 74, 86, 67, and 76. What score does she need on the last test in order to average 80 on her tests? [5pts]

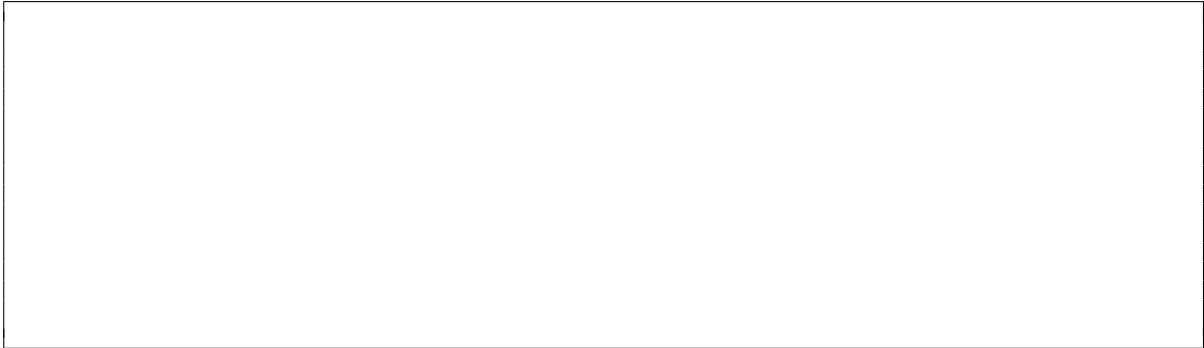
26. Which pair of numbers, when included with the set of data, will raise the value of **both** the mean and median? [5pts]

6, 8, 10, 14, 18

- A. 14 and 15
- B. 1 and 10
- C. 8 and 14
- D. 7 and 15

27. Consider the data set below. Explain how changing one or two of the values can affect the mean. (You may use a calculator.) [10pts]

145, 167, 198, 200, 250

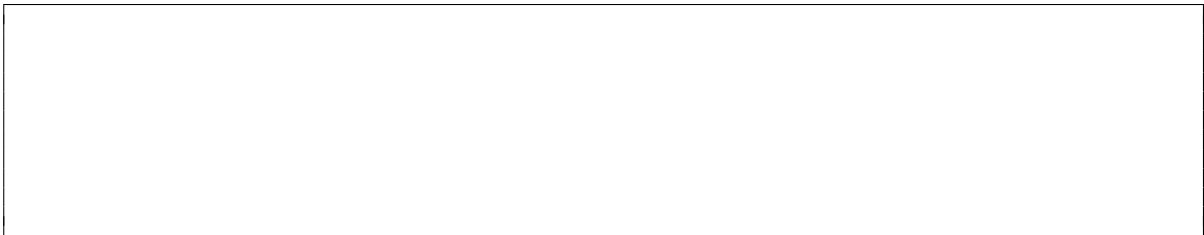


28. Use the strategy *guess and test* to find one data item to add to the data set 9, 12, 12, 14 so that the new set has a mean, median, and mode of 12. [5pts]

29. The Spanish test scores of the students in room 312 are listed below (Jay was absent and took a makeup test later). [10pts]

93, 92, 70, 85, 60, 90, 100, 90, 82, 78, 45, 75,
95, 80, 75, 78, 100, 98, 86

What do we know about Jay's score on his makeup test if the median score on the test (including Jay's score) was 83.5? Explain.



30. The number of patients treated at Dr. Jason's dentist office each day was recorded for eight days. Using the given data, find the mean, median, and mode for this sample: [5pts]

11, 21, 4, 2, 10, 15, 14, 3

- A. mean 10, median 10.5, no mode
 - B. no mean, median 10.5, mode 10
 - C. mean 10.5, median 10, no mode
 - D. mean 21.5, median 10, mode 10.5
31. Carlile collected Food Bank contributions of \$10, \$90, \$40, \$30, and \$10. If he answers the next donor using the *mean* contribution, what amount will he give? [5pts]
- A. \$30
 - B. \$36
 - C. \$90
 - D. \$10
32. The average number of days of thunderstorms at 16 Canadian airports are given. Find the range and the mean. [5pts]

5	10	21	5	22	25	28	28
19	16	2	21	3	26	1	28

- A. (a) range 21; (b) mean 19
 - B. (a) range 23; (b) mean 11.25
 - C. (a) range 28; (b) mean 20
 - D. (a) range 27; (b) mean 16.25
33. The temperature on Mars reaches 27°C during the day and -125°C at night. What is the average temperature? [5pts]
- A. $+76^{\circ}\text{C}$
 - B. -49°C
 - C. -76°C
 - D. $+49^{\circ}\text{C}$

34. Find the mean water temperature for a shallow lake given these readings:

[5pts]

72° , 74° , 71° , 74° , 73° , 74°

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35. Randy recorded the following low temperatures during one week in January:

[5pts]

-5°F , 1°F , 3°F , -10°F , -4°F , -7°F , 12°F .

Use a calculator to find the mean low temperature for that week. Round to the nearest degree.

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36. A teacher asked his class of 20 students, "What is your age?" Their responses are shown on the line plot below.

[10pts]



Find the mean, median, and mode of the data.

Mean: _____

Median: _____

Mode: _____

37. The ages of a group of friends are listed:

[10pts]

48, 32, 29, 33, 52, 48, 45, 33.

Find the mean, median, and mode.

Mean: _____

Median: _____

Mode: _____

38. Marty sent 4 packages. Their masses were 3 kg, 500 g, 660 g, and 5 kg. What was the approximate average mass of the packages?

[5pts]

- A. 2300 g
- B. 5 kg
- C. 6500 g
- D. 3 kg

39. This frequency table shows data from accident reports at a traffic police station.

[10pts]

Length of skid mark (m)	Number of skids
20	13
25	22
30	24
35	44
40	43
45	54

Find the median of the skid mark lengths. Use that value for d in the formula $s = \sqrt{15d}$ to find the corresponding car speed in meters per second, to the nearest whole number.

Median skid length (d , in m): _____

Speed s (m/s, nearest whole number): _____